

Activity-1

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BSCS - 3A

CS311
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Code

Validators.cs

```
1  using System.Collections.Generic;
2  using System.Linq;
3  using Avalonia.Controls;
4  using Avalonia.Input;
5
6  namespace frmPrelimRecitation2;
7
8  public class CleanException : System.Exception
9  {
10     public CleanException(string message) : base(message) { }
11
12     // Overriding this removes the "System.Exception:" prefix when printing the
13     // exception
14     public override string ToString()
15     {
16         return Message;
17     }
18
19     internal class Validators
20     {
21         private HashSet<TextBox> validatedTextBoxes = new HashSet<TextBox>();
22         private string errorMsg = "";
23
24         public string? validateTextBox(in TextBox textBox)
25         {
26             validatedTextBoxes.Add(textBox);
27             if (string.IsNullOrWhiteSpace(textBox.Text))
28             {
29                 errorMsg = $"{textBox.Name} is empty.";
30                 return errorMsg;
31             }
32             else
33             {
34                 return null;
35             }
36         }
37     }
```

```

38
39     public string? keyPressValidator(KeyEventArgs e, in TextBox textBox)
40     {
41         validatedTextBoxes.Add(textBox);
42
43         // 34 in avalonia keycode is '0'
44         int key = (int)e.Key - 34;
45         if (key ≥ 0 && key ≤ 9)
46         {
47             e.Handled = false;
48             return null;
49         }
50         else
51         {
52             e.Handled = true;
53             errorMsg = "Input is not numeric.";
54             return errorMsg;
55         }
56     }
57
58     public int countErrors()
59     {
60         int errorCount = 0;
61         foreach (var textBox in validatedTextBoxes)
62         {
63             errorCount += DataValidationErrors.GetErrors(textBox)?.Count() ?? 0;
64         }
65         return errorCount;
66     }
67
68     public void clearErrors()
69     {
70         foreach (var textBox in validatedTextBoxes)
71         {
72             DataValidationErrors.ClearErrors(textBox);
73         }
74         validatedTextBoxes.Clear();
75     }
76
77 }

```

Lecture 2

- Both inputs are empty, select subtract, and click submit.

Input num1:

txtFirstNumber is empty.

Input num2:

txtSecondNumber is empty.

Select an operation:

☐ Add

☒ Subtract

☐ Multiply

☐ Divide

- Both inputs are not numeric, select subtract, and click submit.

Input num1:

Input is not numeric.

Input num2:

Input is not numeric.

Select an operation:

☐ Add

☒ Subtract

☐ Multiply

☐ Divide

- Both inputs are number, select subtract, and click submit.

Input num1:

Input num2:

Select an operation:

☐ Add

☒ Subtract

☐ Multiply

☐ Divide

The result is: 19

```
1  using Avalonia.Controls;
2  using Avalonia.Input;
3  using Avalonia.Interactivity;
4
5  namespace frmPrelimRecitation2;
6
7  public partial class LectureProgram2 : Window
8  {
9      public LectureProgram2()
10     {
11         InitializeComponent();
12     }
13
14     Validators validators = new Validators();
15     private int num1, num2, result;
16
17     private void btnsubmit_Click(object sender, RoutedEventArgs e)
18     {
19
20         if (validators.validateTextBox(txtFirstNumber) is string s)
21         {
22             DataValidationErrors.SetError(txtFirstNumber, new
23                 CleanException(s));
24         }
25         else
```

```

25     {
26         DataValidationErrors.ClearErrors(txtFirstNumber);
27     }
28
29     if (validators.validateTextBox(txtSecondNumber) is string s2)
30     {
31         DataValidationErrors.SetError(txtSecondNumber, new
            CleanException(s2));
32     }
33     else
34     {
35         DataValidationErrors.ClearErrors(txtSecondNumber);
36     }
37
38     if (validators.countErrors() == 0)
39     {
40         num1 = int.Parse(txtFirstNumber.Text!);
41         num2 = int.Parse(txtSecondNumber.Text!);
42         if (rbAdd.IsChecked == true)
43         {
44             result = num1 + num2;
45         }
46         else if (rbSubtract.IsChecked == true)
47         {
48             result = num1 - num2;
49         }
50         else if (rbMultiply.IsChecked == true)
51         {
52             result = num1 * num2;
53         }
54         else if (rbDivide.IsChecked == true)
55         {
56             result = num1 / num2;
57         }
58         var dialog = new DialogWindow($"The result is: {result}");
59         dialog.ShowDialog(this);
60     }
61
62
63
64     }
65
66     private void txtFirstNumber_KeyDown(object sender, KeyEventArgs e)
67     {
68         if (validators.keyPressValidator(e, txtFirstNumber) is string s)
69         {

```

```

70         DataValidationErrors.SetError(txtFirstNumber, new
           CleanException(s));
71     }
72     else
73     {
74         DataValidationErrors.ClearErrors(txtFirstNumber);
75     }
76 }
77
78 private void txtSecondNumber_KeyDown(object sender, KeyEventArgs e)
79 {
80     if (validators.KeyPressValidator(e, txtSecondNumber) is string s)
81     {
82         DataValidationErrors.SetError(txtSecondNumber, new
           CleanException(s));
83     }
84     else
85     {
86         DataValidationErrors.ClearErrors(txtSecondNumber);
87     }
88 }
89
90 private void btnClear_Click(object sender, RoutedEventArgs e)
91 {
92     validators.clearErrors();
93     txtFirstNumber.Text = "";
94     txtSecondNumber.Text = "";
95     rbAdd.IsChecked = true;
96     txtFirstNumber.Focus();
97 }
98 }

```

Lecture 3

- Input 1 is empty and input2 is a character/string, select subtract and multiply, and click submit.

Input num1:

txtFirstNumber is empty.

Input num2:

Input is not numeric.

Select an operation:

☐ Add

☒ Subtract

☒ Multiply

☐ Divide

- Input 1 is a character/string and input 2 is empty, select subtract and multiply, and click submit.

Input num1:

Input is not numeric.

Input num2:

txtSecondNumber is empty.

Select an operation:

☐ Add

☒ Subtract

☒ Multiply

☐ Divide

- Input 1 is empty and input 2 is a number, select subtract and multiply, and click submit.

Input num1:

txtFirstNumber is empty.

Input num2:

Select an operation:

☐ Add

☒ Subtract

☒ Multiply

☐ Divide

- Both inputs are number, select subtract and multiply, and click submit.

Input num1:

Input num2:

Select an operation:

☐ Add

☒ Subtract

☒ Multiply

☐ Divide

Difference: 9
Product: 736

```
1 using System;  
2 using Avalonia.Controls;
```

C#


```

3  using Avalonia.Input;
4  using Avalonia.Interactivity;
5
6  namespace frmPrelimRecitation2;
7
8  public partial class LectureProgram3 : Window
9  {
10     public LectureProgram3()
11     {
12         InitializeComponent();
13     }
14
15     Validators validators = new Validators();
16     private int num1, num2, result;
17
18     private void btnsubmit_Click(object sender, RoutedEventArgs e)
19     {
20
21         if (validators.validateTextBox(txtFirstNumber) is string s)
22         {
23             DataValidationErrors.SetError(txtFirstNumber, new
24                 CleanException(s));
25         }
26         else
27         {
28             DataValidationErrors.ClearErrors(txtFirstNumber);
29         }
30
31         if (validators.validateTextBox(txtSecondNumber) is string s2)
32         {
33             DataValidationErrors.SetError(txtSecondNumber, new
34                 CleanException(s2));
35         }
36         else
37         {
38             DataValidationErrors.ClearErrors(txtSecondNumber);
39         }
40
41         if (validators.countErrors() == 0)
42         {
43             if (!cbAdd.IsChecked!.Value && !cbSubtract.IsChecked!.Value && !
44                 cbMultiply.IsChecked!.Value && !cbDivide.IsChecked!.Value)
45             {
46                 var errdialog = new DialogWindow("No operation was selected.");
47                 errdialog.ShowDialog(this);
48                 return;
49             }
50         }
51     }
52 }

```

```
47
48     num1 = int.Parse(txtFirstNumber.Text!);
49     num2 = int.Parse(txtSecondNumber.Text!);
50
51     string msg = "";
52     if (cbAdd.IsChecked!.Value)
53     {
54         result = num1 + num2;
55         msg = $"Sum: {result}" + Environment.NewLine;
56     }
57
58     if (cbSubtract.IsChecked!.Value)
59     {
60         result = num1 - num2;
61         msg += $"Difference: {result}"
62             + Environment.NewLine;
63     }
64
65     if (cbMultiply.IsChecked!.Value)
66     {
67         result = num1 * num2;
68         msg += $"Product: {result}"
69             + Environment.NewLine;
70     }
71
72     if (cbDivide.IsChecked!.Value)
73     {
74         result = num1 / num2;
75         msg += $"Quotient: {result}"
76             + Environment.NewLine;
77     }
78     var dialog = new DialogWindow(msg);
79     dialog.ShowDialog(this);
80 }
81 }
82
83 private void txtFirstNumber_KeyDown(object sender, KeyEventArgs e)
84 {
85     if (validators.KeyPressValidator(e, txtFirstNumber) is string s)
86     {
87         DataValidationErrors.SetError(txtFirstNumber, new
            CleanException(s));
88     }
89     else
90     {
91         DataValidationErrors.ClearErrors(txtFirstNumber);
92     }
```

```

93     }
94
95     private void txtSecondNumber_KeyDown(object sender, KeyEventArgs e)
96     {
97         if (validators.KeyPressValidator(e, txtSecondNumber) is string s)
98         {
99             DataValidationErrors.SetError(txtSecondNumber, new
              CleanException(s));
100        }
101        else
102        {
103            DataValidationErrors.ClearErrors(txtSecondNumber);
104        }
105    }
106
107    private void btnclear_Click(object sender, RoutedEventArgs e)
108    {
109        validators.clearErrors();
110        txtFirstNumber.Text = "";
111        txtSecondNumber.Text = "";
112        cbAdd.IsChecked = false;
113        cbSubtract.IsChecked = false;
114        cbMultiply.IsChecked = false;
115        cbDivide.IsChecked = false;
116        txtFirstNumber.Focus();
117    }
118 }

```

Lecture 4

- Input 1 is a number and input 2 is empty, select divide, and click submit.

Input num1:

Input num2:

txtSecondNumber is empty.

Select an operation:

- Input 1 is a character/string and input 2 is a number, select divide, and click submit.

Input num1:

Input is not numeric.

Input num2:

Select an operation:

- Input 1 is a number and input 2 is a character/string, select divide, and click submit.

Input num1:

Input num2:

Input is not numeric.

Select an operation:

- Both inputs are number, select divide, and click submit.

Input num1:

Input num2:

Select an operation:

The result is: 39

```
1 using Avalonia.Controls;  
2 using Avalonia.Input;
```

C#

```

3  using Avalonia.Interactivity;
4
5  namespace frmPrelimRecitation2;
6
7  public partial class LectureProgram4 : Window
8  {
9      public LectureProgram4()
10     {
11         InitializeComponent();
12     }
13
14     Validators validators = new Validators();
15     private int num1, num2, result;
16
17     private void btnsubmit_Click(object sender, RoutedEventArgs e)
18     {
19
20         if (validators.validateTextBox(txtFirstNumber) is string s)
21         {
22             DataValidationErrors.SetError(txtFirstNumber, new
23                 CleanException(s));
24         }
25         else
26         {
27             DataValidationErrors.ClearErrors(txtFirstNumber);
28         }
29
30         if (validators.validateTextBox(txtSecondNumber) is string s2)
31         {
32             DataValidationErrors.SetError(txtSecondNumber, new
33                 CleanException(s2));
34         }
35         else
36         {
37             DataValidationErrors.ClearErrors(txtSecondNumber);
38         }
39
40         if (validators.countErrors() == 0)
41         {
42             num1 = int.Parse(txtFirstNumber.Text!);
43             num2 = int.Parse(txtSecondNumber.Text!);
44
45             if (cmbOperation.SelectedIndex == -1)
46             {
47                 var errdialog = new DialogWindow("Please select an
48                     operation.");
49                 errdialog.ShowDialog(this);

```

```
47         return;
48     }
49
50     if (cmbOperation.SelectedIndex == 0)
51     {
52         result = num1 + num2;
53     }
54     else if (cmbOperation.SelectedIndex == 1)
55     {
56         result = num1 - num2;
57     }
58     else if (cmbOperation.SelectedIndex == 2)
59     {
60         result = num1 * num2;
61     }
62     else if (cmbOperation.SelectedIndex == 3)
63     {
64         result = num1 / num2;
65     }
66     var dialog = new DialogResult($"The result is: {result}");
67     dialog.ShowDialog(this);
68 }
69
70
71
72 }
73
74 private void txtFirstNumber_KeyDown(object sender, KeyEventArgs e)
75 {
76     if (validators.KeyPressValidator(e, txtFirstNumber) is string s)
77     {
78         DataValidationErrors.SetError(txtFirstNumber, new
79             CleanException(s));
80     }
81     else
82     {
83         DataValidationErrors.ClearErrors(txtFirstNumber);
84     }
85 }
86
87 private void txtSecondNumber_KeyDown(object sender, KeyEventArgs e)
88 {
89     if (validators.KeyPressValidator(e, txtSecondNumber) is string s)
90     {
91         DataValidationErrors.SetError(txtSecondNumber, new
92             CleanException(s));
93     }
94 }
```

```
92         else
93         {
94             DataValidationErrors.ClearErrors(txtSecondNumber);
95         }
96     }
97
98     private void btnclear_Click(object sender, RoutedEventArgs e)
99     {
100         validators.clearErrors();
101         txtFirstNumber.Text = "";
102         txtSecondNumber.Text = "";
103         cmbOperation.SelectedIndex = -1;
104         txtFirstNumber.Focus();
105     }
106 }
```

Video

https://drive.google.com/file/d/1Jc-CcFJevc-LhK47GU_NKpSy7lsNFQ5B/view?usp=sharing